

Abiraterone plus prednisone added to androgen deprivation therapy and docetaxel in *de novo* metastatic castration-sensitive prostate cancer: Results from PEACE-1, a multicenter, open label, randomized, phase 3 study with a 2x2 factorial design.

Prof. Karim Fizazi^{1*}, MD, Stéphanie Foulon^{2,3}, MD, Prof. Joan Carles⁴, MD, Guilhem Roubaud⁵, MD, Prof. Ray McDermott⁶, MD, Aude Fléchon⁷, MD, Prof. Bertrand Tombal⁸, MD, Prof. Stéphane Supiot⁹, MD, Dominik Berthold¹⁰, MD, Philippe Ronchin¹¹, MD, Gabriel Kacso¹², MD, Gwenaëlle Gravis¹³, MD, Fabio Calabro¹⁴, MD, Jean-François Berdah¹⁵, MD, Ali Hasbini¹⁶, MD, Marlon Silva¹⁷, MD, Antoine Thiery-Vuillemin¹⁸, MD, Igor Latorzeff¹⁹, MD, Loïc Mourey²⁰, MD, Brigitte Laguerre²¹, MD, Sophie Abadie-Lacourtoisie²², MD, Etienne Martin²³, MD, Claude El Kouri²⁴, MD, Anne Escande²⁵, MD, Alvar Rosello²⁶, MD, Nicolas Magne²⁷, MD, Friederike Schlurmann²⁸, MD, Frank Priou²⁹, MD, Marie-Eve Chand-Fouche³⁰, MD, Salvador Villà Freixa³¹, MD, Muhammad Jamaluddin³², MD, Isabelle Rieger³³, Alberto Bossi³⁴, MD.

Author's affiliation:

¹ Department of Cancer Medicine, Institut Gustave Roussy, University of Paris Saclay, 114 Rue Edouard Vaillant, 94805 Villejuif, France

² Department of Biostatistics and Epidemiology, Gustave Roussy, University Paris-Saclay, Villejuif, France

³ Oncostat U1018, Inserm, University Paris-Saclay, labeled Ligue Contre le Cancer, Villejuif, France

⁴ Vall d'Hebron Institute of Oncology. Vall d'Hebron University Hospital, 08035 Barcelona, Spain

- 1 ⁵ Institut Bergonié 229 cours de l'Argonne, 33076 Bordeaux, France
- 2 ⁶ St. Vincent's University Hospital, Dublin & Cancer Trials Ireland
- 3 ⁷ Centre Léon Bérard, 28 rue Laennec, 69373 Lyon, France
- 4 ⁸ Clinique Universitaire St Luc, Avenue Hippocrate 10, 1200 Brussels, Belgium
- 5 ⁹ Institut de Cancérologie de l'Ouest - site René Gauducheau, Bd Jacques Monod, 44805 Saint-
- 6 Herblain, France
- 7 ¹⁰ Centre Pluridisciplinaire d'Oncologie, Centre Hospitalier Universitaire Vaudois, Rue de Bugnon
- 8 46, 1011 Lausanne, Switzerland
- 9 ¹¹ Centre Azureen de Cancerologie, 1 place du Dr JL Broquerie, 6250 Mougins, France
- 10 ¹² Iuliu Hatieganu University of Medicine and Pharmacy Cluj-Napoca, Amethyst Radiotherapy
- 11 Center, 407280, Cluj, Romania
- 12 ¹³ Institut Paoli-Calmettes, Aix-Marseille Université, CRCM, 232 Boulevard Ste Marguerite,
- 13 13273 Marseille, France
- 14 ¹⁴ San Camillo Forlanini Hospitals, Circonvallazione Gianicolense, 87, 00152 Roma, Italia
- 15 ¹⁵ Clinique Sainte Marguerite, Avenue Victoria, 83400 Hyeres, France
- 16 ¹⁶ Clinique Pasteur, 32 rue Auguste Kervern, 29200 Brest, France
- 17 ¹⁷ Centre François Baclesse, 3 Av du Général Harris, 14076 Caen Cedex 05, France
- 18 ¹⁸ CHU JEAN MINJOZ, 3 Boulevard Alexandre Fleming, 25030 Besancon, France
- 19 ¹⁹ Clinique Pasteur, 1 rue de la petite vitesse BP27617, 31076 Toulouse Cedex3, France
- 20 ²⁰ IUCT – Oncopole, 1 Avenue Irène Joliot Curie, 31059 Toulouse Cedex, France
- 21 ²¹ Centre Eugène Marquis - CS 44 229, 35042 Rennes Cedex, France
- 22 ²² Institut de Cancerologie de l'Ouest, 15 rue Andre Boquel, 49055 Angers cedex 02, France
- 23 ²³ Centre Georges-François Leclerc, 1 rue Professeur Marion, BP 77980, 21079 Dijon, France

- 1 ²⁴ Centre Catherine de Sienne, 2 rue Eric Tabarly - BP 20215, 44202 Nantes Cedex 2, France
- 2 ²⁵ Strasbourg Oncologie Libérale, 184 route de la Wantzenau, 67000 Strasbourg, France
- 3 ²⁶ Institut Català d'Oncologia, Hospital Universitari Josep Trueta, Av Francia s/n, 17007 Girona,
- 4 Spain
- 5 ²⁷ Institut de Cancérologie Lucien Neuwirth, 108 bis avenue albert Raimond, 42271 St Priest en
- 6 Jarez, France
- 7 ²⁸ CHIC Quimper, 14 avenue Yves Thepot, 29107 Quimper, France
- 8 ²⁹ CHD Vendée, Boulevard Stéphane Moreau, 85925 La Roche sur Yon, France
- 9 ³⁰ Centre Antoine Lacassagne, 33 Avenue de Valombrese, 06189 Nice, France
- 10 ³¹ Institut Català d'Oncologia, Cap de Servei Oncologia Radioteràpica, Hospital Universitari
- 11 Germans Trias, Ctra. de Canyet s/n, 08916 Badalona, Catalunya, Spain
- 12 ³² Cork University Hospital, Wilton Road, T12 DC4A Cork, Ireland
- 13 ³³ Unicancer, 101 rue de Tolbiac, 75654 Paris cedex 13, France
- 14 ³⁴ Department of Radiotherapy, Institut Gustave Roussy, University of Paris Saclay, 114 Rue
- 15 Edouard Vaillant, 94805 Villejuif, France
- 16
- 17 ***Corresponding author:** Karim Fizazi, MD, PhD, Department of Cancer Medicine, Institut
- 18 Gustave Roussy, Paris Saclay University, 114 Rue Edouard Vaillant, 94805 Villejuif, France.
- 19 Tel. : +33 (0)1 42 11 43 17
- 20 Fax : +33 (0) 1 42 11 52 11
- 21 Email : karim.fizazi@gustaveroussy.fr

Abstract

Background: Current standards of care (SOC) for metastatic castration-sensitive prostate cancer (mCSPC) supplement androgen-deprivation therapy (ADT) with either docetaxel, second-generation hormonal therapy, or radiotherapy. PEACE-1 appraised whether combining those supplements would prompt superior outcomes.

Methods: The active, open-label, 2x2 factorial designed, phase III, PEACE-1 (ClinicalTrials.gov: NCT01957436) conducted at 77 sites across seven European countries equally randomized *de novo* mCSPC patients to SOC (ADT alone or with docetaxel), SOC+abiraterone (1000 mg abiraterone acetate plus 10 mg prednisone, daily), SOC+radiotherapy, or SOC+radiotherapy+abiraterone. The co-primary endpoints were radiographic progression-free survival (rPFS) and overall survival (OS). Abiraterone efficacy was first assessed in the overall population then in the ADT+docetaxel population (population of interest).

Findings: The 1173 patients included between November 27, 2013 and December 20, 2018 were followed for medians of 3.5 and 4.4 years for rPFS and OS, respectively. Adjusted Cox regression modelling revealed no interaction between abiraterone and radiotherapy, enabling the pooled analysis of abiraterone efficacy. In the overall population, patients allocated to abiraterone (n=583) had longer rPFS (Hazard Ratio [HR], 0.54; 99.9% confidence interval [CI], 0.41 to 0.71; $p<0.001$) and OS (HR, 0.82; 95.1% CI, 0.69 to 0.98; $p=0.03$) compared to SOC (n=589). In the ADT+docetaxel population (n=355 in both group), adding abiraterone improved rPFS (HR, 0.50; 99.9% CI, 0.35 to 0.72, $p<0.001$) and OS (HR, 0.75; 95.1% CI, 0.59 to 0.95, $p=0.02$). In the ADT+docetaxel population, grade 3-5 adverse events increased from 52% to 63% by adding

abiraterone, with hypertension being the most impacted (from 13% to 22%). The triplet combination did not increase the rates of neutropenia, febrile neutropenia, fatigue, and neuropathy.

Interpretation: Combining ADT, docetaxel and abiraterone for *de novo* mCSPC improves survival with a modest increase in significant toxicity, mostly hypertension.

Funding: This work was supported by Janssen-Cilag, Ipsen, Sanofi, and the French government PHRC-K-13-050 and PRT-K20-116 grants.

Introduction

After several decades with no significant progress, the treatment of *de novo* (synchronous) metastatic prostate cancer has drastically evolved in recent years with the addition of diverse treatments to androgen deprivation therapy (ADT), the former standard of care (SOC). Hence, the prognosis of *de novo* metastatic castration-sensitive prostate cancer (mCSPC) men has been improved chronologically by using ADT concomitantly with either docetaxel¹⁻⁵ or one of the second-generation androgen receptor axis inhibitors abiraterone acetate,⁶⁻⁸ apalutamide,⁹ or enzalutamide.^{10,11} In 2018, radiotherapy to the primary tumor was also shown to extend the survival of patients with low-volume metastatic disease.¹² The benefits of these various combinatory therapies were confirmed by meta-analyses¹³⁻¹⁵ and have shaped the current guidelines.¹⁶⁻¹⁸

Nevertheless, whether merging some of these new treatments could provide further clinical benefits and if so, what combination may be most suitable for *de novo* metastatic castration-sensitive prostate cancer (mCSPC) remains to be established.¹⁹ Conducted by a European

consortium (PEACE),²⁰ the international, phase III trial PEACE-1 relies upon a 2x2 factorial design to assess whether abiraterone acetate combined with prednisone (abiraterone) and/or local radiotherapy to the primary tumor should be combined with SOC (ADT+/-docetaxel). Here, we report on the primary endpoint results regarding the efficacy of abiraterone on radiographic progression-free survival (rPFS) and overall survival (OS) in the overall population and in the ADT+docetaxel population.

Methods

Study design

PEACE-1 is an open-label, randomized, active-controlled phase III trial with a 2x2 factorial design conducted at 77 sites across seven European countries (Belgium 54 (5%) patients, France 920 (79%), Ireland 60 (5%), Italy, Romania, Spain, and Switzerland). The initial protocol was reviewed and first approved on July 10th, 2013 by the French Independent Ethics Committee (CPP Ile de France VII). The protocol complied with the Declaration of Helsinki's ethical principles and was approved by Institutional Review Boards at each site. The trial conformed with the International Conference on Harmonization and Good Clinical Practice guidelines, and applicable regulatory requirements. Steering and Independent Data Monitoring Committees (IDMC) roles and members are described in supplementary appendix 2.1.

Patients

The complete list of inclusion and non-inclusion criteria is provided as supplemental material. Briefly, patients over 18 years with histologically or cytologically confirmed prostate adenocarcinomas documented as *de novo* metastatic by bone scan, CT scan, or MRI (according to

Response Evaluation Criteria in Solid Tumors [RECIST] version 1.1 criteria) and an Eastern Cooperative Oncology Group (ECOG) performance status score of 0–2 were eligible for inclusion provided they had received an ADT no more than 3 months ahead of randomization with at least 6 weeks separating the initiation of ADT from the first docetaxel dose. Patients presenting with a pure small-cell carcinoma, or prior prostate cancer treated by a definitive local treatment were not included. All participants gave written informed consent before any study specific procedures or randomization were initiated.

Modification of the protocol during the study

Two major amendments were made during the accrual period to take into account the evolution of SOC for men with mCSPC. While ADT alone was the only SOC when the trial began accrual in 2013, the trial was amended on October 5, 2015 to allow for docetaxel use after ADT+docetaxel had demonstrated OS improvement^{2,3} (273 patients had already been accrued). Then, in 2017 abiraterone was also shown to improve OS compared to ADT alone.^{6,7} Thereby, an additional amendment was filed (August 10, 2017) to make docetaxel mandatory for the rest of the patients to be accrued, so that the only question about systemic treatment the trial would then randomly address was that of the addition of abiraterone to ADT+docetaxel (given that the question of ADT+abiraterone compared to of ADT alone was already addressed by other trials). Trial amendments are summarized in the Appendix. Evolution of the sample size and patients' allocation according to these key amendments is depicted in Figure S1.

Sample size

The initial protocol of 2013 planned a sample size of 916 patients. However, due to SOC evolution and the knowledge gained from the LATITUDE and STAMPEDE trials results published in late 2017, the sample size was increased to 1173 and the co-primary objectives were revised (see Appendix section 5). Since PEACE-1 was the only phase III trial that could answer the question of abiraterone efficacy in mCSPC men treated with ADT+docetaxel as SOC, the timing of the final analyses was based on this subgroup. Sample size determination was performed with the East[®] software (Cytel, MA, USA) based on the assumption that no statistically significant interaction would take place between radiotherapy and abiraterone to allow for a 2x2 factorial analysis of the abiraterone efficacy. For abiraterone effect evaluation the predetermined acceptable probability of committing a type I error was set at 0.05 divided between the two co-primary endpoints: OS (0.049) and rPFS (0.001). As hypothesis, adding abiraterone to ADT+docetaxel was to improve OS by 30% and rPFS by 40% over median times of 53 and 30 months, respectively. Hence, with 355 patients allocated to ADT+docetaxel (+/-radiotherapy)+abiraterone and 355 patients allocated to ADT+docetaxel (+/- radiotherapy), 249 death events would give an 80% power to detect an OS hazard ratio (HR) of 0.70 at a two-sided alpha level of 0.049. With the same allocation, 262 radiographic progression or death events were predicted to have an 80% power to detect a rPFS HR of 0.60 at a two-sided alpha level of 0.001.

Randomization and masking

Eligible patients were centrally randomized from November 2013 to December 2018 over the “Alea Clinical Portal” (<https://www.aleaclinical.eu>) to equally allocated patients in a 1:1:1:1 ratio (Figure 1). This randomization process was performed via the Tenalea autonomous software solely

accessed by the trial data manager within the Epidemiology and Biostatistics facility of the Gustave Roussy Center in Villejuif - France up to 2015, then by each investigator.

The randomized allocation was done by means of a minimization algorithm taking into account the enrollment center, the ECOG performance status (0 vs. 1-2), the type of ADT (GnRH agonist vs. antagonist vs. bilateral orchiectomy), the planned administration of docetaxel (yes vs. no), and the disease extent/burden based on metastatic status (lymph nodes only vs. bone [with or without lymph nodes] vs. the presence of visceral metastases). A similar weight was given to each of the five stratification factors with a 0.8 probability to minimize imbalance in the number of patients assigned to each treatment. As the study's case report form gathered information for patient stratification according to high vs. low volume disease burden classification as defined in the CHAARTED study² [with high volume being defined as the presence of visceral metastases or at least four bone lesions with one beyond the vertebral bodies and pelvis], this later classification was retained to favor inter-trial comparison. No masking was performed during this open-label study.

Procedures

The treatment flow chart is described in Figure S2. Patients allocated to docetaxel with continuous ADT were to receive six cycles of 75 mg/m²/cycle (maximum dose of 150 mg/cycle) of docetaxel to be administered every 3 weeks with a +/- 3 days tolerance. Granulocyte colony-stimulating factor (G-CSF) injections after each docetaxel cycle were recommended until the January 22, 2018 amendment made G-CSF prophylaxis mandatory for docetaxel-treated patients. The first docetaxel cycle had to take place within 14 days after randomization and be distant from ADT initiation by at least 6 weeks. Those assigned to "abiraterone" received abiraterone acetate plus prednisone:

1 1000 mg of abiraterone acetate (four 250 mg tablets, orally), administered once daily along with
2 prednisone 5 mg orally twice daily within 6 weeks of ADT initiation. PEACE-1 started accrual in
3 2013, far before the LATITUDE and the STAMPEDE-Abiraterone trials read out and obviously
4 before abiraterone was approved in the mCSPC indication. This is why we used 10 mg of
5 prednisone in PEACE-1, similar to what was and is approved for mCRPC. “Abiraterone” was
6 administered until progression to castration-resistance, consent withdrawal, unacceptable toxicity,
7 or death. ADT was continuously maintained by either a GnRH agonist, a GnRH antagonist, or
8 bilateral orchiectomy. Patients assigned to radiotherapy (74 Gray in 37 fractions distributed over
9 7 to 8 weeks) should have completed their docetaxel regimen by at least 3 and no more than 8
10 weeks. Treatments given after progression followed common practice with **investigators being at**
11 **liberty to change** therapy, as they thought was most appropriate, based on PSA level variation or
12 clinical progression even in absence of radiographic evidence being allowed.

13 Every enrolled patient will be followed for a total duration of ten years. The complete list of
14 assessments performed at each visit and follow-up intervals are detailed in Table S1. The vital
15 status of 97% of the patients was gathered within 3 months of the cutoff date.

17 **Outcomes**

18 The co-primary endpoints were rPFS and OS. Radiographic progression of soft-tissue lesions was
19 evaluated by either CT or MRI based on RECIST v1.1. Progression of bone lesions was assessed
20 by bone scan according to the adapted version of Prostate Cancer Working Group 2 criteria⁶ with
21 no secondary bone scan being required to confirm a progression. rPFS was defined as the time
22 between randomization and the occurrence of radiographic progression or death from any cause.

OS was the time between randomization and death from any cause. Patients without events were censored at the date of last follow-up.

The secondary outcomes were Castration-Resistant Prostate Cancer Free Survival (CRPC-FS), Serious Genitourinary event-free survival (S-GU-EFS), prostate cancer-specific survival, time to next skeletal-related event, PSA response rate, prognostic study of serum PSA measured 6-8 months after initiation of systemic therapy, time to pain progression, time to chemotherapy for CRPC, Quality of life, changes in bone mineral density (BMD), correlation of biomarkers with outcome, event rate per 100-subject years of treatment analysis, and toxicity. CRPC-FS represented the time between randomization and CRPC or death from any cause. CRPC was defined by either radiographic progression or a confirmed PSA rise (based on 3 independent measurements A, B and C with $A < B < C$, and $C \geq 0.50$ ng/mL), with a serum testosterone within castrated range (< 0.50 ng/mL). Adverse events grading was based on the National Cancer Institute Common Terminology Criteria for Adverse Events version 4.0 (NCI-CTCAE v4) classification. Safety analysis was based on the highest grade recorded during the period spanning from treatment initiation to CRPC. Prostate cancer-specific survival corresponded to the time between randomization and the occurrence of death from prostate cancer. The event rate per 100-subject years of treatment was the number of adverse events divided by the amount of subject-time observed (i.e., 100 subjects experiencing any adverse event over a 1-year period of exposure, with an exposure time going from randomization to CRPC or last follow-up). Apart from CRPC-FS, cancer-specific survival, event rate per 100-subject years and toxicity, secondary outcomes results were still under investigation at the time this manuscript was submitted.

Statistical analysis

PEACE-1 trial has a factorial design, based on the assumption that no interaction between abiraterone and radiotherapy would take place. Before analyzing the various outcomes, the presence of such an interaction for both co-primary endpoints was tested by analyses of maximum likelihood estimates for a Cox model adjusting on stratification factors (ECOG performance status [0 vs. 1-2], ADT type [GnRH agonist vs. GnRH antagonist vs. bilateral orchiectomy], disease burden² [low vs. high, see “Randomization and masking” section for definition], docetaxel [no docetaxel pre- vs. post-amendment vs. with docetaxel post-amendment]). In absence of a qualitative interaction (p -value >0.05), the arms were to be combined two by two based on abiraterone administration regardless of the radiotherapy treatment before sequentially assessing abiraterone efficacy first in the overall population, then in the ADT+docetaxel population (assuming the significance level was reached in first instance) (Figure 1). Of note, as the preplanned number of rPFS and OS events has not yet been reached in the population of mCSPC men presenting with a low-volume disease, the efficacy of radiotherapy will be analyzed in due course (Supplementary appendix chapter 7).

All efficacy analyses were conducted on the intent-to-treat population (ITT), i.e. patients were analyzed according to the treatment group assigned by randomisation (Figure 1). Safety analyses were conducted on the safety population, excluding patients who did not receive any investigational treatment and analyzing patients according to the treatment group they actually received.

The median follow-up was estimated by the inverse Kaplan-Meier method. Efficacy analyses were done on the intent to treat population. Time to event endpoints were estimated by the Kaplan-Meier method. Cox proportional-hazards model adjusted on radiotherapy and stratification factors provided statistical significances and an estimate of the abiraterone effect (p -value and HR with

confidence intervals [CI] adjusted to match adjustment made to significance levels in the corresponding test [i.e., 99.9% for rPFS; 95.1% for OS; 95% for CRPC-FS]). Non parametric confidence interval for the median survival differences between groups were analyzed by the *bootkm* function of the R *Hmisc* package (5000 repeats). Statistical analyses were performed with SAS® 9.4 (SAS, NC, USA) and R 4.0.2.

Role of the funding sources

None of the financial providers was involved in the study design, data collection, data analysis, data interpretation, or writing of the manuscript.

Results

Between November 27, 2013 and December 20, 2018, 1173 patients were equally randomized to SOC only, SOC+abiraterone, SOC+radiotherapy, or SOC+radiotherapy+abiraterone. To reflect clinical guidelines evolution over the years, amendments to the protocol were authorized (see supplementary appendix 5 and Figure S1) so that SOC was either ADT (+/- radiotherapy) or ADT+docetaxel (+/- radiotherapy) for 462 and 710 patients, respectively. Patient allocation is depicted in Figure 1. One patient withdrew his consent and his data and therefore the study population comprises 1172 patients. The matching baseline characteristics of the populations analyzed are presented in Table 1. Briefly, the median age was 67 years with 57% and 43% of the patients having a high and low volume disease,² respectively. One patient withdrew his consent and therefore the study population comprises 1172 patients.

The co-primary endpoint results presented here correspond to the final planned analysis. For disease progression and overall survival, the median follow-up periods were longer in the overall

1 population (3.5 and 4.4 years) than in the ADT+docetaxel population (3.0 and 3.8 years).
2 Regardless of the population and survival outcomes considered, the median follow-up periods
3 were not different between the groups receiving or not abiraterone. For rPFS analyses (Figure 2A
4 and 2B), the median follow-up periods from randomization to the September 1st, 2020 cutoff were
5 3.53 years for the overall population (3.52 years for SOC and 3.56 years for SOC+Abi, log-rank
6 test $p=0.86$) and 2.99 years for the ADT+docetaxel population (3.00 years for SOC and 2.97 years
7 for SOC+Abi, log-rank test $p=0.93$). For OS analyses (Figure 2C, 2D, 2E, and 2F), the median
8 follow-up periods from randomization to the June 1st, 2021 cutoff were 4.41 years for the overall
9 population (4.44 years for SOC and 4.39 years for SOC+Abi, log-rank test $p=0.98$) and 3.81 years
10 for the ADT + docetaxel population (3.75 years for SOC and 3.85 years for SOC+Abi, log-rank
11 test $p=0.95$). No interim analysis was conducted.

12 Due to the 2x2 factorial design of the trial, we sought to test the presence of a statistical interaction
13 between abiraterone and radiotherapy on the co-primary (rPFS and OS), and secondary (CRPC-
14 FS) efficacy outcomes. Within the overall population ($n=1172$), no interaction between abiraterone
15 and radiotherapy for either rPFS ($p=0.64$), OS ($p=0.86$), CRPC-FS ($p=0.56$), or cancer-specific
16 survival ($p=0.54$) were found adjusting on four stratification factors. Likewise, no interactions
17 were found within the docetaxel population ($n=710$) for either rPFS ($p=0.94$), OS ($p=0.85$),
18 CRPC-FS ($p=0.75$), or cancer-specific survival ($p=0.98$) adjusting on ECOG performance status,
19 ADT type and disease burden. Consequently, the comparative evaluation of abiraterone efficacy
20 on survival outcomes was conducted by pooling the arms two by two (Arms A+C vs. Arms B+D).
21 In the overall population, the addition of abiraterone decreased the number of radiographic
22 progression or death events from 371 to 252, increased the median rPFS from 2.22 to 4.46 years

1 and reduced by 46% the relative risk of radiographic progression or death compared to SOC alone
2 (adjusted hazard ratio [HR], 0.54; 99.9% confidence interval [CI], 0.41 to 0.71; $p<0.001$) (Figure
3 2A). Notably, the addition of abiraterone to SOC (ADT +/- docetaxel +/- radiotherapy) lowered
4 the number of deaths from 268 to 228, improved the median OS from 4.72 to 5.72 years and
5 reduced by 18% the risk of death from any cause (adjusted HR, 0.82; 95.1% CI, 0.69 to 0.98;
6 $p=0.03$) (Figure 2C). In the overall population the effect of abiraterone on rPFS and OS was
7 evaluated according to the predefined subgroups and demonstrated a high consistency across most
8 of them but the underrepresented surgically castrated patients and those deprived of docetaxel
9 based on investigators decision. For OS only, the effect of abiraterone supplementation was more
10 pronounced in men with high-volume (Figure S3).

11 As SOC+abiraterone demonstrated a superior efficacy compared to SOC alone that met the
12 required predefined significance level within the overall population (type one error of 0.049 for
13 OS and 0.001 for rPFS), abiraterone efficacy was further investigated within the more restricted
14 ADT+docetaxel population to assess the pre-planned co-primary endpoints (Figure 1). Compared
15 to ADT+docetaxel (+/-radiotherapy) alone, the addition of abiraterone decreased the number of
16 radiographic progression or death events from 211 to 139, increased the median rPFS from 2.03
17 to 4.46 years and reduced by 50% the relative risk of radiographic progression or death (adjusted
18 HR, 0.50; 99.9% CI, 0.35 to 0.72; $p<0.001$) (Figure 2B and 3A). Similarly, compared to
19 ADT+docetaxel (+/-radiotherapy) alone, the addition of abiraterone reduced the number of deaths
20 from 151 to 121, extended the median OS from 4.43 years to a not yet reached point while lowering
21 by 25% the risk of death from any cause (adjusted HR, 0.75; 95.1% CI, 0.59 to 0.95; $p=0.02$)
22 (Figure 2D).

The ADT+docetaxel population was further analyzed based on disease burden. Compared to ADT+docetaxel (+/-radiotherapy) alone, the addition of abiraterone decreased the number of radiographic progression or death events from 55 to 41, and from 156 to 97, and reduced by 42% and 53% the relative risk of radiographic progression or death in low- and high-volume patients, respectively (low-volume: adjusted HR, 0.58; 99.9% CI, 0.29-1.15; $p=0.006$; medians: not reached [3.1 and not reached] vs. 2.7 years [2.5 to not reached]; high-volume: adjusted HR, 0.47; 99.9% CI, 0.30 to 0.72; $p<0.001$; medians: 4.1 [2.7 to not reached] vs. 1.6 years [1.4 to 2.0]) (Figure S4). Median OS in high-volume disease patients rose from 3.47 to 5.14 years when abiraterone was added, corresponding to a 28% lower risk of death (adjusted HR, 0.72; 95.1% CI, 0.55 to 0.95; $p=0.02$) (Figure 2F). As the data were not mature, no conclusive response could be obtained for low-volume disease patients (Figure 2E and 3B).

Adding abiraterone to ADT+docetaxel (+/-radiotherapy) delayed castration resistance (HR, 0.38; 95% CI, 0.31 to 0.47; $p<0.001$; medians, 1.45 vs. 3.21 years) (Figure S5). Notably, among the ADT+docetaxel patients who developed castration resistance and were still alive at the cutoff date, 221 of 263 (84%) were subsequently treated by at least one life-prolonging therapy and 213 of 263 (81%) by at least one next-generation hormonal therapy in the control arm, compared to 104 of 141 (74%) and 65 of 141 (46%) in the abiraterone arm, respectively. See Table S2 for a description of the treatments prescribed post progression. Cancer-specific survival was also improved by the addition of abiraterone to ADT+docetaxel (HR, 0.69; (95% CI 0.53 to 0.90; $p=0.006$; medians, 4.72 vs. not reached) (Figure S6).

Adding abiraterone to ADT+docetaxel had no impact on the number of docetaxel cycles administered (median, 6; interquartile range [IQR], 6-6 in both groups). On the other hand, 138 of 226 (61%) vs. 183 of 347 (53%) patients were discontinued from abiraterone administration,

including 29 of 138 (21%) vs. 32 of 183 (17%) due to toxicity in the no docetaxel vs. docetaxel-treated subgroups, respectively (Table S3). Overall, the median abiraterone treatment duration before discontinuation was 16.5 months (95% CI, 8.7 to 31.0) vs. 15.1 months (95% CI, 7.9 to 25.1) among no docetaxel and docetaxel-treated patients, respectively.

In the ADT+docetaxel safety population treated or not with abiraterone, 217 of 347 patients (63%) vs. 181 of 350 patients (52%) patients experienced at least one severe adverse event (Grade 3-5) and 7 (2%) vs. 3 (1%) patients experienced a fatal adverse event (Table 2 and Table S4). However, 217 of 347 (63%) vs. 149 of 226 (66%) patients treated by abiraterone were affected by at least one severe adverse event and 7 (2%) vs. 8 (4%) patients experienced a fatal adverse event in the ADT+docetaxel population and ADT population, respectively (Table 2), indicating that the addition of docetaxel to abiraterone did not increase the incidence of severe or lethal adverse events. Accordingly, 40 vs. 55 severe adverse events per 100-subject years were recorded in the ADT+docetaxel receiving and deprived of abiraterone, respectively (Table S5). The incidence of frequent severe adverse events (Grade 3-5 adverse events reported in 5% or more of the patients) was similar between abiraterone-treated and abiraterone-deprived patients in the ADT+docetaxel population, except for hypertension (76 of 347 [22%] vs. 45 of 350 [13%]) and hepatotoxicity with increased transaminases (20 of 347 [6%] vs. 2 of 350 [$<1\%$]) which were more frequent among abiraterone recipients (Table 2). While, the incidence of severe (Grade 3-5) hypertension and hepatotoxicity were similar in response to abiraterone among ADT+docetaxel and ADT populations, a marked opposite was observed as expected for neutropenia which affected 34 of 347 (10%) vs. 0 of 226 (0%) patients in those respective populations (Table S6).

Even though severe (Grade 3-5) fatigue and peripheral neuropathy events did not reach the frequency cutoff of 5% in the ADT+docetaxel population, 10 of 347 (3%) vs. 15 of 350 (4%)

patients experienced at least one count of severe fatigue, while 4 of 347 (1%) vs. 6 of 350 (2%) experienced at least one severe peripheral neuropathy event when treated or deprived of abiraterone, respectively (Table 2).

Discussion

To the best of our knowledge PEACE-1 is the first trial to demonstrate that a triple systemic therapy consisting of ADT, docetaxel and a second-generation androgen signaling inhibitor (abiraterone) extends rPFS (HR, 0.50; 99.9% CI, 0.35 to 0.72; $p<0.001$; medians, 2.03 and 4.46 years) and OS (HR, 0.75; 95.1% CI, 0.59 to 0.96; $p=0.02$; medians: 4.43 and not reached) in patients with *de novo* mCSPC.

Whether the systemic treatment of high-volume disease (i.e., extended metastatic spread) should differ from the one of low-volume disease has been extensively debated in recent years. The benefit of ADT+docetaxel was clearly demonstrated for mCSPC patients with high-volume disease,^{4,21} despite data collected on larger populations support a similar benefit across high- and low-volume disease.³ This discrepancy may result from different ratios of patients with *de novo* vs. relapsing low-volume disease included in these trials since such recurrences favor better prognoses.²² On the other hand, second-generation androgen receptor axis inhibitors prompt OS benefits in mCSPC men regardless of the volume considered.^{5,9,10} The mature PEACE- 1 data for men with high-volume disease indicate a clear improvement in both rPFS (HR, 0.47; 99.9% CI, 0.30 to 0.72; $p<0.001$; medians: 1.62 and 4.05 years) and OS (HR, 0.72; 95.1% CI, 0.55 to 0.95; $p=0.02$; medians: 3.47 vs 5.14 years) when abiraterone is added to ADT+docetaxel. Even though OS data were immature for low-volume disease, an abiraterone-mediated rPFS benefit was already apparent in those men (HR, 0.58; 99.9% CI, 0.29 to 1.15; $p=0.006$; medians: 2.71 years vs not

reached) (Table 3). Thus, ADT+docetaxel+abiraterone should be proposed as a new SOC for men fit for docetaxel prescription presenting with a high-volume disease. However, for those with a low-volume disease, the decision to use two or three agents should be carefully considered by both the patients and their oncologists to determine whether a major rPFS benefit (HR=0.58) is sufficient to justify a treatment intensification in a context of immature OS data. It would also depend on whether or not an OS benefit is required. Long term data and consensus conferences will allow for clinical practice guidance refinement.

As abiraterone can inhibit CYP3A4 *in vitro*,²³ and CYP3A4 is involved in docetaxel elimination,²⁴ the concomitant administration of these two drugs could potentially increase docetaxel exposure and thereby its toxicity. However, no rise in chemotherapy-related side effects occurred in reaction to this concomitance, which may suggest an absence of significant drug-drug interactions between abiraterone and docetaxel. This is consistent with the observation that abiraterone and docetaxel have comparable pharmacokinetic behavior when administered alone or in combination.²⁵ Similarly, abiraterone did not affect the clearance of another taxane, cabazitaxel, by CYP3A.²⁶ We did not either observe an increase of hematologic toxicity when abiraterone was added to ADT+docetaxel in opposition to a phase II trial conducted in mCRPC.²⁷ The most frequent high-grade adverse event reported among abiraterone recipients in PEACE-1 was hypertension (22% vs. 13%). This expected adverse event was carefully monitored and managed to prevent the onset of cardiovascular events, which were not seen in excess in PEACE-1. In the ENZAMET trial, hypertension was also twice more frequent among mCSPC men when enzalutamide was added to ADT+docetaxel.¹⁰ However, with enzalutamide addition a higher incidence of fatigue, syncope and therapy discontinuation was reported, while the rise of such events was not observed in PEACE-1 when abiraterone was added to ADT+docetaxel. Transaminase increase, a common side

1 effect of abiraterone, was indeed observed in PEACE-1 (6% vs 1%), but at a similar level with or
2 without docetaxel. This increase had no detectable clinical consequences due to per recommended
3 management.²⁸ Notably, the addition of abiraterone to ADT+docetaxel did not impact on
4 neutropenia incidence (10% vs. 9%). The impact of toxicity on participants will be better
5 understood after the formal analysis of quality-of-life data, at which time the triplet therapy
6 oncologic benefits may still outweigh the adverse impact of the additional toxicity.

7 Aside the limitations inherently associated with open-labelled trials, the rapid clinical practice
8 evolution for mCSPC that took place during the conduct of this trial prompted the sequential
9 amendment of PEACE-1 protocol and statistical analysis plan to reflect those changes, answer
10 meaningful questions and implement the best treatment options to all participants (appendix 5.1).

11 In particular, as a survival benefit for mCSPC men treated with abiraterone combined with ADT
12 alone was demonstrated in 2017, accrual was then restricted to men receiving ADT+docetaxel as
13 SOC, and our pre-established statistical analysis was revised to specifically assess the efficacy of
14 abiraterone in term of survival outcomes in the ADT+docetaxel population. De facto, further
15 allocating men to ADT only would have been unethical given that the ADT+abiraterone
16 combination was already known to improve survival.

17 PEACE-1 demonstrates that combining abiraterone with ADT+docetaxel improves both rPFS and
18 OS over ADT+docetaxel alone. Indeed, docetaxel and abiraterone have distinct mechanisms of
19 action. In addition, they do not display absolute cross-resistance as abiraterone improves survival
20 after docetaxel treatment and docetaxel retains some activity after abiraterone failure.^{29,30}
21 Collectively, these observations provide a rationale for the survival benefits resulting from their
22 combination evidenced in PEACE-1. However, this study did not answer whether this triple
23 combination yields clinical advantages over ADT plus a second-generation androgen receptor axis

1 inhibitor. Indeed, at the time PEACE-1 was conducted (2013-2018), combining ADT with such
2 an inhibitor had not been approved for mCSPC, thereby precluding such a comparison within the
3 trial. Incidentally, the ongoing phase III ARASENS and ENZAMET trials (ClinicalTrials.gov n°
4 NCT02799602 and NCT02446405) are also testing the combined efficacy of second-generation
5 androgen receptor axis inhibitors (darolutamide and enzalutamide, respectively) with
6 ADT+docetaxel and their upcoming OS results will put abiraterone's benefits in perspective.

7 Achieving treatments' synergy and preventing acquired resistance in spite of tumor clonal
8 heterogeneity are cardinal aspects of mCSPC management requiring an optimization of the
9 treatment paradigm. In PEACE-1, substantial survival benefits were obtained when abiraterone
10 was added to ADT+docetaxel. Remarkably, these benefits were observed even though more than
11 80% of the control group eventually received a second-generation androgen receptor axis inhibitor
12 (mostly abiraterone or enzalutamide) to address cancer progression. Indeed, in contrast to previous
13 trials, PEACE-1 investigators were allowed to prescribe early salvage treatments to the control
14 group patients presenting with evidence of cancer progression. To this end, a PSA minimum value
15 of just 0.50 ng/mL was used to define mCRPC even in absence of radiographic progression.
16 Collectively, these observations suggest that the upfront combination implemented in PEACE-1
17 is superior to the sequential ordonnance of these (or similar) agents administered as control. Of
18 note, abiraterone acetate is approved for metastatic prostate cancer in many countries and is close
19 to be a worldwide generic drug.

20 In conclusion, PEACE-1 not only confirms that combining abiraterone to SOC improves outcomes
21 in men with *de novo* mCSPC, but it demonstrates that combining abiraterone with ADT and
22 docetaxel improves survival and other important efficacy endpoints with limited significant
23 toxicity increase. Nonetheless, as PEACE-1 only included patients with *de novo* mCSPC, it is still

unclear whether this triple therapy may benefit to metachronous mCSPC patients. Our trial cannot directly address whether this triplet systemic combination is superior to ADT and abiraterone. A longer follow-up is mandated to answer whether combining such an intensive first line systemic treatment with radiotherapy to the primary cancer may bring further clinical benefits to mCSPC patients. This upcoming analysis will be performed when the preplanned number of rPFS and OS events is reached in the population of men presenting with low-volume metastatic dissemination.

Acknowledgements

We deeply appreciate the guidance provided by the members of the Independent Steering Committee and the Independent Data Monitoring Committee and are thankful to all patients who agreed to participate in this study. We also appreciate the help provided by members of the Department of Biostatistics and Epidemiology at Gustave Roussy. Namely, we would like to thank Nathalie Bouvet-Forteau for setting-up the randomization process as well as Matthieu Texier and Aurélie Bardet for providing interim safety statistical analyses to the Independent Data Monitoring Committee meetings as well as Marie-Laure Tanguy and Agnès Laplanche for their statistical inputs. We are also grateful for the help of Amel Belhimeur from the Institut du Cancer de Montpellier for timely data management, and the assistance of Hélène Ribault, Muriel Habibian, Jean-Carlos Cascais, Steeve Castiglione, Sadène Tezkratt, Dorsafe Fekiri, Tri Hahn Dung Doan and Ange Grigbale regarding trial management at Unicancer under the direction of Soazig Nenan. We thank the European Organisation for Research and Treatment of Cancer (EORTC). We express our gratitude for the financial support provided by Janssen-Cilag, Ipsen and Sanofi. We would also like to thank Sebastien Marion, Medical Writer at Unicancer for writing support and figure design.

References

1. Gravis G, Fizazi K, Joly F, et al. Androgen-deprivation therapy alone or with docetaxel in non-castrate metastatic prostate cancer (GETUG-AFU 15): a randomised, open-label, phase 3 trial. *Lancet Oncol* 2013; **14**(2): 149-58.
2. Sweeney CJ, Chen YH, Carducci M, et al. Chemohormonal Therapy in Metastatic Hormone-Sensitive Prostate Cancer. *N Engl J Med* 2015; **373**(8): 737-46.
3. James ND, Sydes MR, Clarke NW, et al. Addition of docetaxel, zoledronic acid, or both to first-line long-term hormone therapy in prostate cancer (STAMPEDE): survival results from an adaptive, multiarm, multistage, platform randomised controlled trial. *Lancet* 2016; **387**(10024): 1163-77.
4. Kyriakopoulos CE, Chen YH, Carducci MA, et al. Chemohormonal Therapy in Metastatic Hormone-Sensitive Prostate Cancer: Long-Term Survival Analysis of the Randomized Phase III E3805 CHAARTED Trial. *J Clin Oncol* 2018; **36**(11): 1080-7.
5. Clarke NW, Ali A, Ingleby FC, et al. Addition of docetaxel to hormonal therapy in low- and high-burden metastatic hormone sensitive prostate cancer: long-term survival results from the STAMPEDE trial. *Ann Oncol* 2019; **30**(12): 1992-2003.
6. Fizazi K, Tran N, Fein L, et al. Abiraterone plus Prednisone in Metastatic, Castration-Sensitive Prostate Cancer. *N Engl J Med* 2017; **377**(4): 352-60.
7. James ND, de Bono JS, Spears MR, et al. Abiraterone for Prostate Cancer Not Previously Treated with Hormone Therapy. *N Engl J Med* 2017; **377**(4): 338-51.
8. Fizazi K, Tran N, Fein L, et al. Abiraterone acetate plus prednisone in patients with newly diagnosed high-risk metastatic castration-sensitive prostate cancer (LATITUDE): final overall survival analysis of a randomised, double-blind, phase 3 trial. *Lancet Oncol* 2019; **20**(5): 686-700.
9. Chi KN, Chowdhury S, Bjartell A, et al. Apalutamide in Patients With Metastatic Castration-Sensitive Prostate Cancer: Final Survival Analysis of the Randomized, Double-Blind, Phase III TITAN Study. *J Clin Oncol* 2021; **39**(20): 2294-303.
10. Davis ID, Martin AJ, Stockler MR, et al. Enzalutamide with Standard First-Line Therapy in Metastatic Prostate Cancer. *N Engl J Med* 2019; **381**(2): 121-31.
11. Armstrong A.J. IT, Azad A.A., Szmulewitz R.Z., Holzbeierlein J., Villers A., Alcaraz A., Alekseev B.Y., Shore N.D., Petrylak D.P., Rosbrook B., Yamada F. a, S., Haas G.P., Stenzl A. Final overall survival (OS) analysis from ARCHES: A phase III, randomized, double-blind, placebo (PBO)-controlled study of enzalutamide (ENZA) + androgen deprivation therapy (ADT) in men with metastatic hormone-sensitive prostate cancer (mHSPC). *Annals of Oncology (2021) 32 (suppl_5): S1283-S1346* 101016/annonc/annonc741 2021.
12. Parker CC, James ND, Brawley CD, et al. Radiotherapy to the primary tumour for newly diagnosed, metastatic prostate cancer (STAMPEDE): a randomised controlled phase 3 trial. *Lancet* 2018; **392**(10162): 2353-66.
13. Rydzewska LHM, Burdett S, Vale CL, et al. Adding abiraterone to androgen deprivation therapy in men with metastatic hormone-sensitive prostate cancer: A systematic review and meta-analysis. *Eur J Cancer* 2017; **84**: 88-101.
14. Vale CL, Fisher DJ, White IR, et al. What is the optimal systemic treatment of men with metastatic, hormone-naïve prostate cancer? A STOPCAP systematic review and network meta-analysis. *Ann Oncol* 2018; **29**(5): 1249-57.
15. Feyerabend S, Saad F, Li T, et al. Survival benefit, disease progression and quality-of-life outcomes of abiraterone acetate plus prednisone versus docetaxel in metastatic hormone-sensitive prostate cancer: A network meta-analysis. *Eur J Cancer* 2018; **103**: 78-87.
16. Parker C, Castro E, Fizazi K, et al. Prostate cancer: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. *Ann Oncol* 2020; **31**(9): 1119-34.

17. Schaeffer E, Srinivas S, Antonarakis ES, et al. NCCN Guidelines Insights: Prostate Cancer, Version 1.2021. *J Natl Compr Canc Netw* 2021; **19**(2): 134-43.
18. Cornford P, van den Bergh RCN, Briers E, et al. EAU-EANM-ESTRO-ESUR-SIOG Guidelines on Prostate Cancer. Part II—2020 Update: Treatment of Relapsing and Metastatic Prostate Cancer. *European Urology* 2021; **79**(2): 263-82.
19. Vogl UM, Beer TM, Davis ID, et al. Lack of consensus identifies important areas for future clinical research: Advanced Prostate Cancer Consensus Conference (APCCC) 2019 findings. *European Journal of Cancer* 2021.
20. Fizazi K, Abrahamsson PA, Ahlgren G, et al. Achievements and perspectives in prostate cancer phase 3 trials from genitourinary research groups in Europe: introducing the Prostate Cancer Consortium in Europe. *Eur Urol* 2015; **67**(5): 904-12.
21. Gravis G, Boher J-M, Chen Y-H, et al. Burden of Metastatic Castrate Naive Prostate Cancer Patients, to Identify Men More Likely to Benefit from Early Docetaxel: Further Analyses of CHAARTED and GETUG-AFU15 Studies. *European Urology* 2018; **73**(6): 847-55.
22. Francini E, Gray KP, Xie W, et al. Time of metastatic disease presentation and volume of disease are prognostic for metastatic hormone sensitive prostate cancer (mHSPC). *Prostate* 2018; **78**(12): 889-95.
23. Deb S, Chin MY, Adomat H, Guns ES. Abiraterone inhibits 1 α ,25-dihydroxyvitamin D3 metabolism by CYP3A4 in human liver and intestine in vitro. *J Steroid Biochem Mol Biol* 2014; **144 Pt A**: 50-8.
24. van Waterschoot RA, Lagas JS, Wagenaar E, et al. Absence of both cytochrome P450 3A and P-glycoprotein dramatically increases docetaxel oral bioavailability and risk of intestinal toxicity. *Cancer Res* 2009; **69**(23): 8996-9002.
25. Tagawa ST, Posadas EM, Bruce J, et al. Phase 1b Study of Abiraterone Acetate Plus Prednisone and Docetaxel in Patients with Metastatic Castration-resistant Prostate Cancer. *Eur Urol* 2016; **70**(5): 718-21.
26. Massard C, Mateo J, Lortol Y, et al. Phase I/II trial of cabazitaxel plus abiraterone in patients with metastatic castration-resistant prostate cancer (mCRPC) progressing after docetaxel and abiraterone. *Ann Oncol* 2017; **28**(1): 90-5.
27. Duran MAC, Font A, Duran I, et al. Randomized phase II study of docetaxel (D) + abiraterone•acetate (AA) versus D after disease progression to first-line AA in metastatic castration-resistant prostate cancer (mCRPC): ABIDO-SOGUG Trial. *Journal of Clinical Oncology* 2020; **38**(6_suppl):95-.
28. Colomba E, Marret G, Baciarello G, et al. Liver tests increase on abiraterone acetate in men with metastatic prostate cancer: Natural history, management and outcome. *Eur J Cancer* 2020; **129**: 117-22.
29. Fizazi K, Scher HI, Molina A, et al. Abiraterone acetate for treatment of metastatic castration-resistant prostate cancer: final overall survival analysis of the COU-AA-301 randomised, double-blind, placebo-controlled phase 3 study. *Lancet Oncol* 2012; **13**(10): 983-92.
30. de Bono JS, Smith MR, Saad F, et al. Subsequent Chemotherapy and Treatment Patterns After Abiraterone Acetate in Patients with Metastatic Castration-resistant Prostate Cancer: Post Hoc Analysis of COU-AA-302. *Eur Urol* 2017; **71**(4): 656-64.

Figure legends

Figure 1. Study flow diagram. Abi stands for abiraterone, ADT for androgen-deprivation therapy, DXL for docetaxel, and RT for radiotherapy.

Figure 2. Kaplan-Meier estimates of the radiographic PFS and Overall Survival within the overall and ADT+docetaxel populations. Time to event curves are presented for the overall (A and C) and docetaxel populations (B, D, E and F). SOC is ADT+/-Docetaxel+/-RT for the overall population and ADT+Docetaxel+/-RT for the ADT+Docetaxel population. A and B show rPFS, and C-F show OS.

Figure 3. Hazard ratios for radiographic progression-free survival (A) and overall survival (B) by predefined stratification factors in the ADT+docetaxel population. Hazard ratios are plotted on a linear scale. At time of rPFS analysis (A), disease burden data were not available for six patients which were *de facto* excluded from the Cox model. OS analysis per predefined stratification factor was conducted on the ITT population (B). The number of events per patients in the radiotherapy subgroup analysis is not presented as the efficacy of radiotherapy is still under investigation. ECOG PS denotes Eastern Cooperative Oncology Group (ECOG) Performance Status, ADT Androgen Deprivation Therapy, GnRH Gonadotropin Releasing Hormone (also known as LHRH), SOC Standard of Care.

1 **Table 1. Characteristics of the intent to treat populations at baseline.**

	Treatment allocation			
	Overall population (n=1172)		ADT+DXL population (n=710)*	
	SOC = ADT +/- DXL +/- RT		SOC = ADT + DXL +/- RT	
	SOC + Abi	SOC	SOC + Abi	SOC
	Arm B+D (n=583)	Arm A+C (n=589)	Arm B+D (n=355)	Arm A+C (n=355)
Treated by Radiotherapy	291 (50)	293 (50)	178 (50)	177 (50)
Country				
Belgium	29 (5)	25 (4)	16 (5)	16 (5)
France	458 (79)	462 (78)	278 (78)	280 (79)
Ireland	30 (5)	30 (5)	17 (5)	13 (4)
Italy	1 (<1)	3 (<1)	0 (0)	0 (0)
Romania	4 (<1)	5 (<1)	0 (0)	0 (0)
Spain	55 (9)	56 (10)	38 (11)	39 (11)
Switzerland	6 (1)	8 (1)	6 (2)	7 (2)
Age (years)				
Median	67 (61-72)	66 (59-72)	66 (60-70)	66 (59-70)
Range	37 to 94	43 to 87	37 to 85	44 to 84
ECOG performance status				
0	412 (71)	412 (70)	250 (70)	246 (69)
1 - 2	171 (29)	177 (30)	105 (30)	109 (31)
T stage				
T1	23 (4)	23 (4)	10 (3)	13 (4)
T2	109 (19)	94 (16)	64 (19)	45 (13)
T3	287 (51)	310 (53)	167 (49)	189 (55)
T4	98 (17)	99 (17)	68 (20)	65 (19)

Tx	45 (8)	54 (9)	32 (9)	35 (10)
Missing data	21 (4)	9 (2)	14 (4)	8 (2)
N stage				
N+	307 (55)	325 (57)	198 (58)	207 (60)
N-	186 (33)	174 (30)	99 (29)	97 (28)
Nx	69 (12)	76 (13)	43 (13)	39 (11)
Missing data	21 (4)	14 (2)	15 (4)	12 (3)
Time from diagnosis (months)				
Median	2·3 (1·6-3·2)	2·3 (1·4-3·1)	2·2 (1·6-3·0)	2·2 (1·4-2·9)
Missing data	10 (2)	10 (2)	6 (2)	7 (2)
Metastatic localization				
Bone ^f	472 (81)	475 (81)	287 (81)	279 (79)
Lymph node only	47 (8)	52 (9)	27 (8)	29 (8)
Visceral [†]	64 (11)	62 (11)	41 (12)	47 (13)
Metastatic burden[§]				
High burden	331 (57)	336 (57)	224 (63)	232 (65)
Low burden	252 (43)	253 (43)	131 (37)	123 (35)
Gleason score				
≤ 7	145 (25)	133 (23)	79 (23)	71 (20)
8 - 10	429 (75)	441 (77)	270 (77)	276 (80)
Missing data	9 (2)	15 (3)	6 (2)	8 (2)
PSA at randomization (ng/mL)				
Median	14 (3-62)	11 (3-55)	14 (2-59)	12 (3-63)
Missing data	2 (<1)	4 (<1)	0 (0)	2 (<1)
Medical history				
Hypertension	270 of 574 (47)	241 of 562 (43)	156 of 352 (44)	148 of 344 (43)
Diabetes mellitus type 2	62 of 566 (11)	80 of 556 (14)	33 of 351 (9)	56 of 344 (16)
High cholesterol	229 of 568 (40)	229 of 556 (41)	136 of 351 (39)	130 of 343 (38)

1 Data are n (%) or median (IQR). All numbers are rounded to the nearest integer.

2 * The median number of docetaxel cycles was 6 (interquartile range Q3-Q1,6-6) in both arms.

3 ^f Without visceral.

4 [†] With or without lymph node and bone.

5 [§] The metastatic burden was classified as reported by Sweeney *et al.* 2015 with a high burden

6 characterized by four or more bone metastases with one or more outside the vertebral

7 bodies or pelvis, or visceral metastases, or both; all other assessable situations being classified as

8 low burden.

9 Ethnicity-related information are not presented as French laws forbid the collection of such data.

10 Abi stands for abiraterone, ADT for androgen-deprivation therapy, DXL for docetaxel, RT for

11 radiotherapy, and SOC stands for standard of care.

Table 2. Adverse Events Recorded within the Safety Population Treated or not with Abiraterone with or without Docetaxel.

	Treatment allocation			
	ADT+DXL population		ADT population	
	(SOC = ADT + DXL +/- RT)		(SOC = ADT +/- RT)	
	SOC + Abi	SOC	SOC + Abi	SOC
	Arm B+D (n=347)	Arm A+C (n=350)	Arm B+D (n=226)	Arm A+C (n=237)
Adverse events				
Any adverse events	346 (100)	349 (100)	226 (100)	233 (99)
Severe (Grade 3-5) adverse events	217 (63)	181 (52)	149 (66)	97 (41)
Lethal (Grade 5) adverse events	7 (2)	3 (1)	8 (4)	5 (2)
Frequent severe adverse events per P.T.				
Hypertension	76 (22)	45 (13)	66 (29)	38 (16)
Neutropenia	34 (10)	32 (9)	0 (0)	0 (0)
Hepatotoxicity	20 (6)	2 (1)	14 (6)	3 (1)
Febrile neutropenia	18 (5)	19 (5)	2 (1)	1 (0)
Gamma-glutamyl transferase increased	17 (5)	14 (4)	6 (3)	4 (2)
Erectile dysfunction	7 (2)	5 (1)	12 (5)	13 (5)
Blood alkaline phosphatase increased	15 (4)	12 (3)	6 (3)	13 (5)
Other severe adverse events per P.T.				
Fatigue	10 (3)	15 (4)	3 (1)	0 (0)
Peripheral neuropathy	4 (1)	6 (2)	1 (0)	0 (0)

Data are n (%). As the patients were not randomized according to docetaxel prescription, toxicities recorded in the ADT and ADT+Docetaxel populations are not directly comparable. Percentages are rounded to the nearest integer. The total number of patient (n) corresponds to the ones who actually received abiraterone and/or docetaxel. Severe adverse events (grade 3-5) recorded in >5% of the patients in either group are considered frequent and reported as frequent severe adverse events in decreasing order of occurrence according to Medical Dictionary for Regulatory Affairs (MedDRA) Preferred Term (P.T.) classification.

Abi stands for abiraterone, ADT for androgen-deprivation therapy, DXL for docetaxel, RT for radiotherapy, and SOC stands for standard of care.

1 **Table 3. Abiraterone-mediated efficacy outcomes at a glance – ITT population.**

3a	Overall population (Abi vs. SOC)	Survival medians (yr.) (Abi vs. SOC)	Survival median differences (yr.) (Abi vs. SOC)	HR (CI)	p-value
Primary outcomes					
OS	583 vs. 589	5.7 vs 4.7	0.9 (95.1% CI, 0.0 to 2.0)	0.82 (95.1% CI, 0.69 to 0.98)	0.03
rPFS	583 vs. 589	4.5 vs 2.2	2.1 (99.9% CI, 0.7 to 2.9)	0.54 (99.9% CI, 0.41 to 0.71)	<0.001
Secondary outcomes					
CRPC-FS	583 vs. 589	3.8 vs. 1.5	2.3 (95% CI, 1.6 to 3.0)	0.40 (95% CI, 0.35 to 0.47)	<0.001
Cancer-specific survival	583 vs. 589	NR vs. 5.8	NA	0.75 (95% CI, 0.61 to 0.91)	0.004

3b	Docetaxel population (Abi vs. SOC)	Survival medians (yr.) (Abi vs. SOC)	Survival median differences (yr.) (Abi vs. SOC)	HR (CI)	p-value
Primary outcomes					
OS	355 vs. 355	NR vs 4.4	NA	0.75 (95.1% CI, 0.59 to 0.95)	0.02
rPFS	355 vs. 355	4.5 vs 2.0	2.2 (99.9% CI, 0.7 to 2.8)	0.50 (99.9% CI, 0.35 to 0.72)	<0.001
Secondary outcomes					
OS	Low-volume (131 vs. 123)	NR vs. NR	NA	0.83 (95.1% CI, 0.50 to 1.39)	0.66
OS	High-Volume (224 vs. 232)	5.1 vs 3.5	1.1 (95.1% CI, 0.2 to 1.9)	0.72 (95.1% CI, 0.55 to 0.95)	0.02
rPFS	Low-volume (129 vs. 122)	NR vs. 2.7	NA	0.58 (99.9% CI, 0.29 to 1.5)	0.006
rPFS	High-Volume (225 vs. 231)	4.1 vs 1.6	2.2 (99.9% CI, 0.6 to 3.2)	0.47 (99.9% CI, 0.30 to 0.72)	<0.001
CRPC-FS	355 vs. 355	3.2 vs 1.4	1.8 (95% CI, 1.5 to 3.1)	0.38 (95% CI, 0.31 to 0.47)	<0.001
Cancer-specific survival	355 vs. 355	NR vs. 4.7	NA	0.69 (95% CI, 0.53 to 0.90)	0.006

2

3 Primary and secondary outcomes are displayed for the Overall population (**3a**) and the Docetaxel population (**3b**). NR stands for not

4 reached, NA for non-available.

Contribution statement:

KF, AB and SF conceived and designed the study. KF, JC, GR, RMD, AF, BT, SS; DB; PR; GK; GG; FC; JFB, AH, MS, ATV, IL, LM, BL, SAL, EM, CEK, AE, AR, NM, FS, FP, MECF, SVF, MJ and AB were investigators who conducted the study and collected data. KF, IR and SF directly accessed and verified the underlying data. KF, JC, GR, RMD, DB, FB, BT, GK, GG, ATV and SF contributed to data analysis or data interpretation.

All authors participated in data interpretation, manuscript writing, review, and approval of the final version of the manuscript for submission.

Data sharing statement

The data collected and analyzed for the purpose of the present manuscript are not immediately available due to ethical and legal restrictions. However, Unicancer will grant access to all de-identified individual data underlying the published results upon written and detailed request originating from all personal involved in cancer research sent to peace1@unicancer.fr.

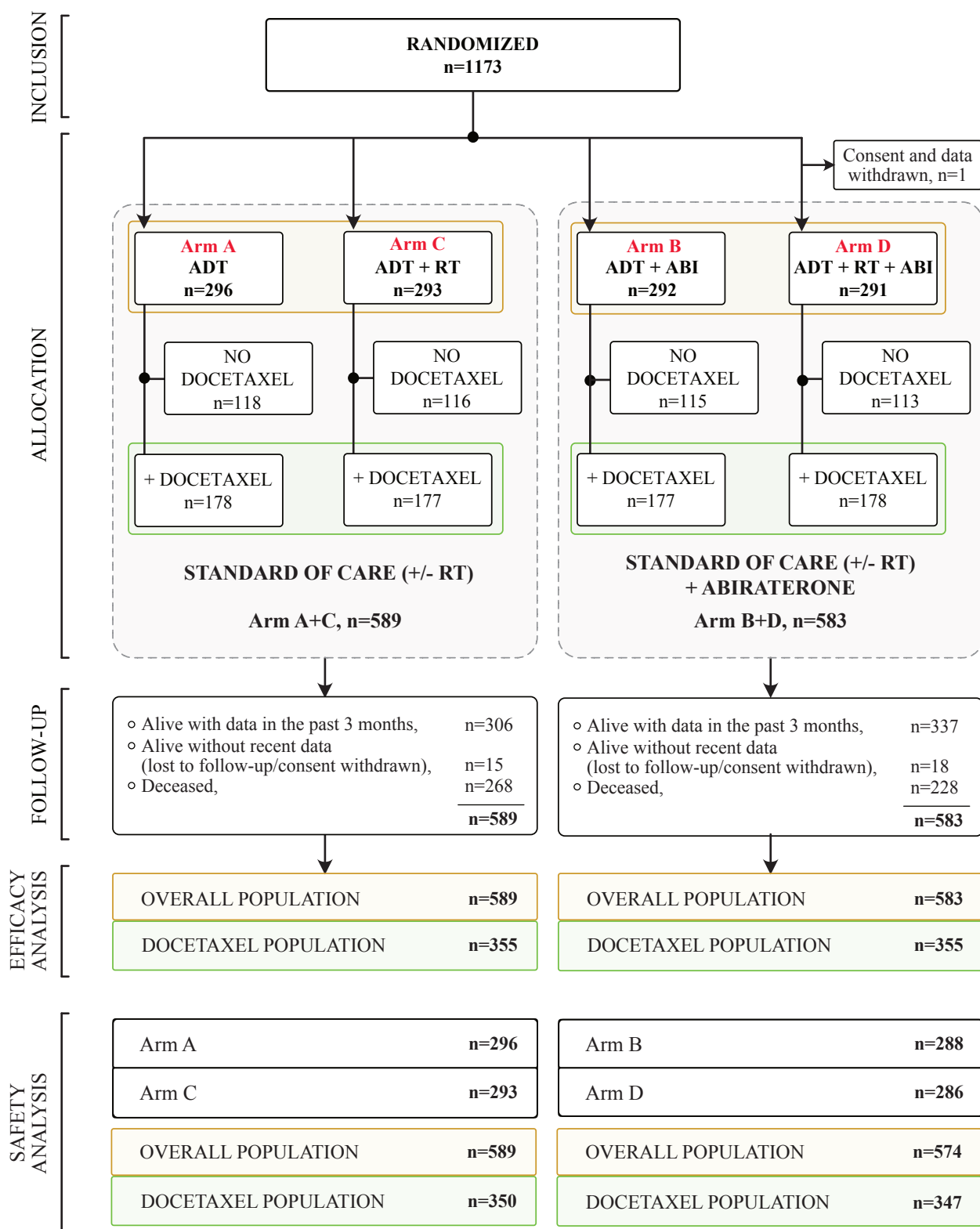


Figure 1

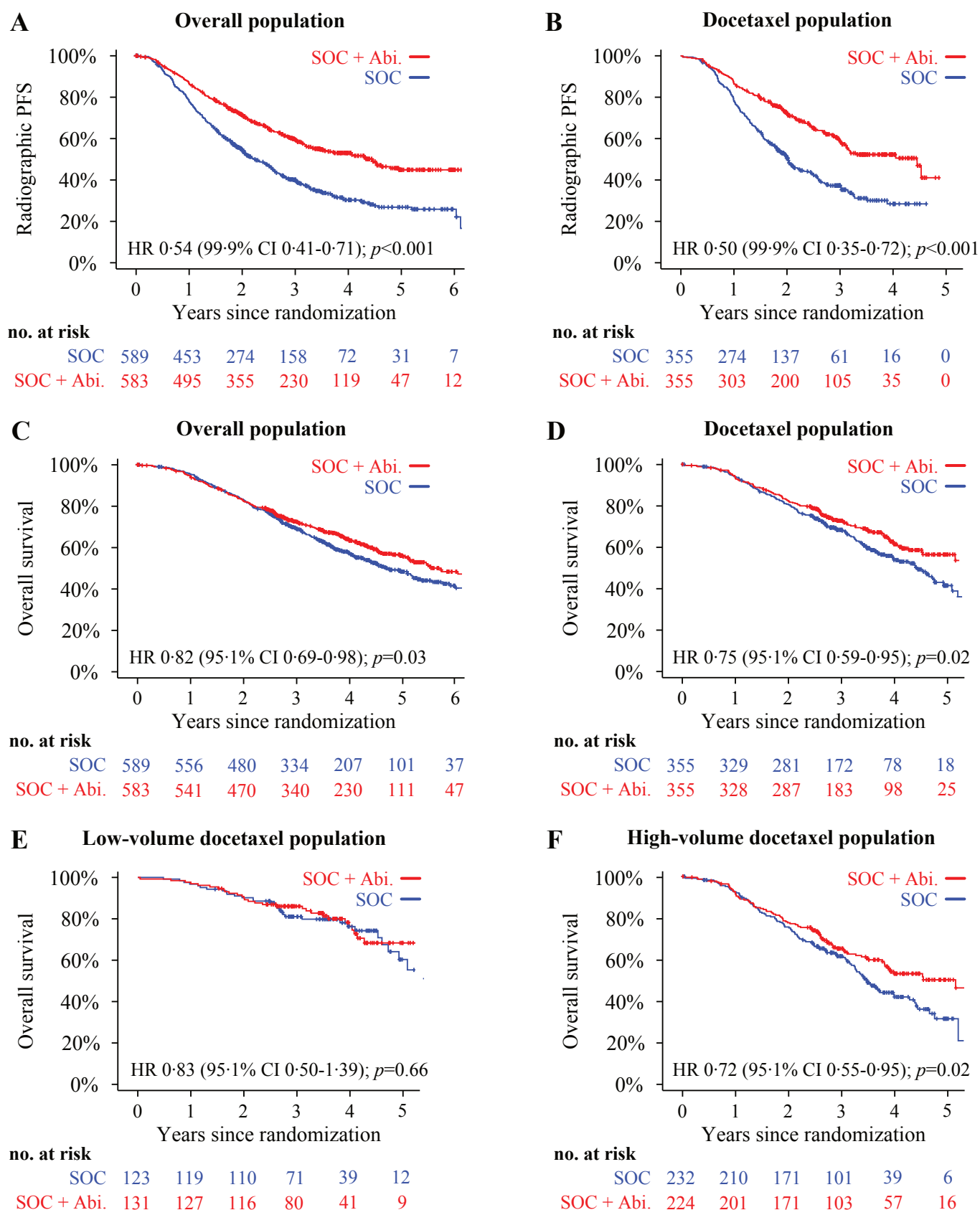
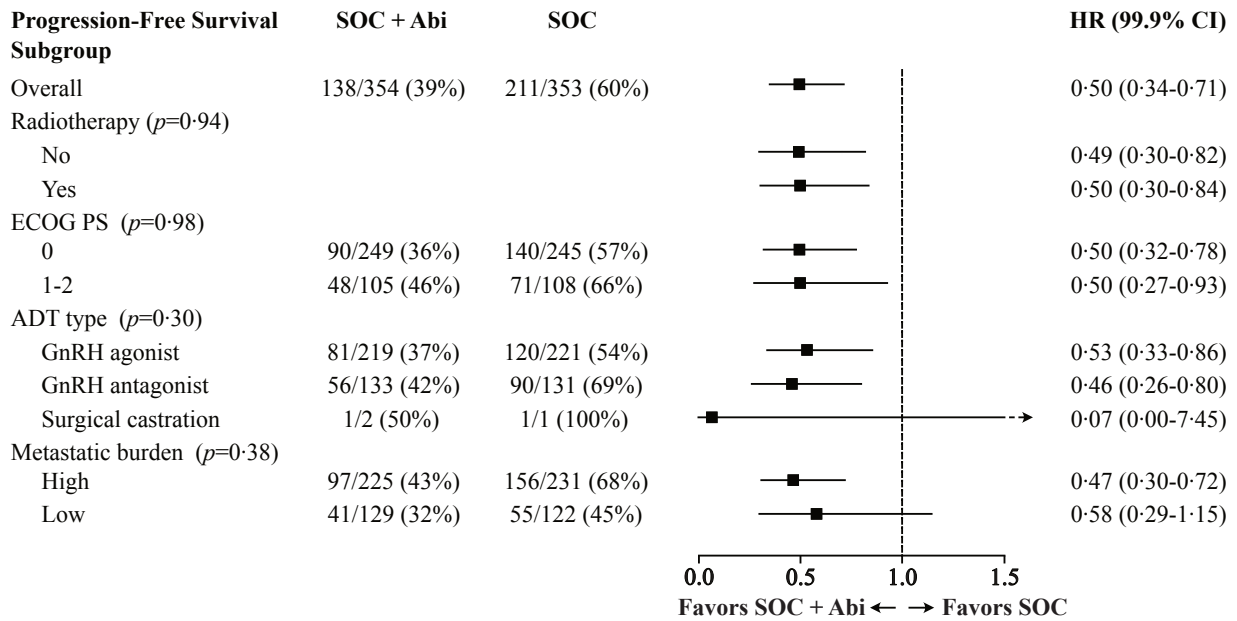


Figure 2

A



B

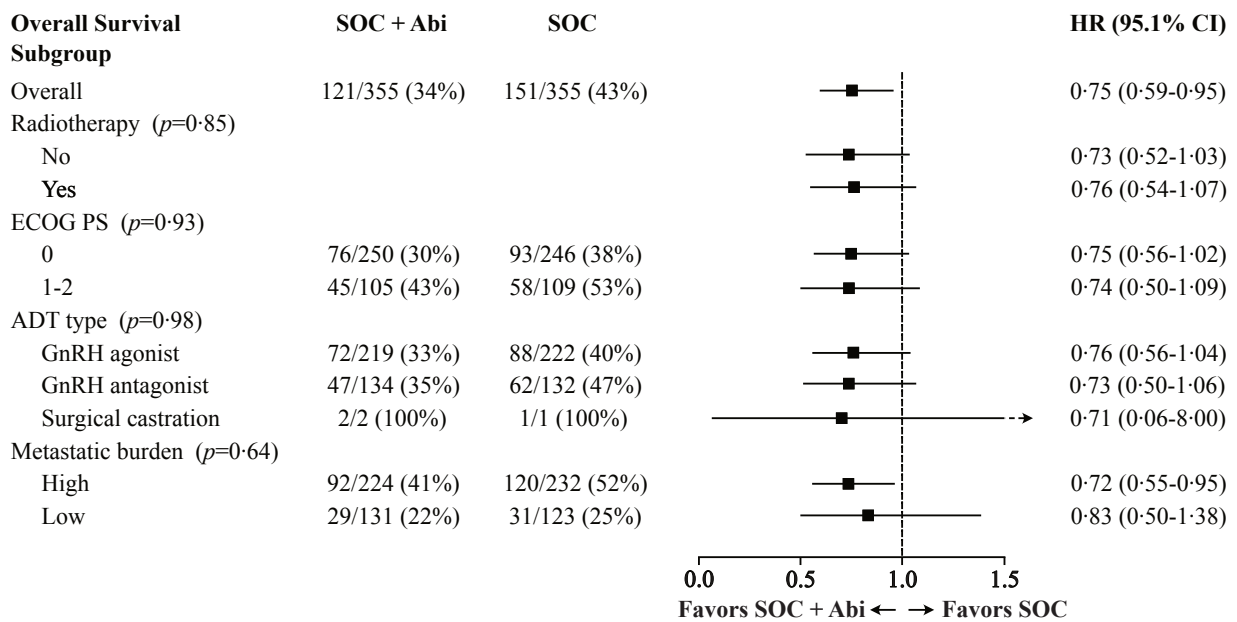


Figure 3